Imprinted Proteins as a Receptor in Fluorescent Sensing Microplate Assay for Herbicides Determination

Supplementary material

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Table **S1**: Maximum permissible levels of imazamox in cereal plants in different countries

Country	Culture	Concentration limit	Ref.
Russia	Sunflower seeds	0.004 $mg\,m^{-3}$	[1]
USA	Wheat	0.3 ppm	[2]
China	Wheat	0.05 $mg\,kg^{-1}$	[3]
Australia	Barley	0.02–0.04 $mg\,kg^{-1}$	[4]
EU	Sunflower seeds	0.30 $mg\,kg^{-1}$	[5]

[1]Federal Center for Hygiene and Epidemiology of Rospotrebnadzor. Guidelines for control methods. Collection. MUK 4.1.2665—10
[2]Federal Register. Vol. 66, No. 248. Rules and Regulations 66773
[3]National Food Safety Standard—Maximum Residue Limit of Pesticides in Foods (GB 2763–2021)
[4]Public release summary on “Evaluation of the new active imazamox in the products raptor herbicide & raptor wg herbicid". National Registration Authority for Agricultural and Veterinary Chemicals. NRA Refs: 50853 & 50854
[5]Commission Regulation (EU) 2021/2202 of 9 December 2021

Table **S2**: Molecularly imprinted polymers selective to herbicides

Monomer / Cross-linked agent	Detectable substances	Substrate	LOD	Linear range	Recovery	Ref.
1–VN	Imazapyr	n/a	0.09 μg L^{-1}	0.29–200.0 μg L^{-1}	86–107 %	[1]
	Imazapic	=	0.06 μg L^{-1}	0.21–200.0 μg L^{-1}	=	
	Imazethapyr	=	0.04 μg L^{-1}	0.15–200.0 μg L^{-1}	=	
	=	fCBPE	0.03 μmol L^{-1}	0.10–70.00 μmol L^{-1}	96.3–105.7 %	[2]
AA / TRIM	=	CPR	15 ng g^{-1}	0.1–5.0 μg mL^{-1}	91.1–97.5 %	[3]
	=	Fe_3O_4@SiO_2	2.13 μg L^{-1}	5–100 μg L^{-1}	87.7–102 %	[4]
4–VP / EGDMA	=	SPME fiber	0.16 μg mL^{-1}	0.50–50 μg mL^{-1}	87.5–123 %	[5]
	Imazameth	=	0.16 μg mL^{-1}	=	70.0–99.9 %	
	Imazamox	=	0.070 μg mL^{-1}	=	60.0–87.9 %	
	Imazapyr acid	=	0.16 μg mL^{-1}	=	63.1–87.0 %	
	Imazaquin acid	=	0.29 μg mL^{-1}	1.0–100 μg mL^{-1}	81.2–99.5 %	

All papers in the table used **imazethapyr** as a template.

List of acronyms: **=** —similar value, **1–VN**—1-vinylimidazole, **4–VP**—4-vinylpyridine, **AA**—acrylamide, **CPR**—chloromethylation polystyrene resin, **EGDMA**—ethylene glycol dimethacrylate, **fCBPE**—functionalized carbon black paste electrode, **LOD**—limit of detection, **MA**—methacrylic acid, **SPME**—solid phase microextraction, **TRIM**—trimethylolpropane trimethacrylate

[1] doi.org/10.1016/j.foodchem.2020.127345—Development of selective preconcentration/clean-up method for imidazolinone herbicides determination in natural water and rice samples by HPLC-PAD using an imazethapyr imprinted poly(vinylimidazole-TRIM)

[2] doi.org/10.1002/elan.202100360—Development of a Novel Molecularly Imprinted Polyvinylimidazole/Functionalized Carbon Black Nanocomposite-based Paste Electrode for Electrochemical Sensing of Imazethapyr in Rice Samples

[3] doi.org/10.1155/2018/7535417—Preparation of Imazethapyr Surface Molecularly Imprinted Polymers for Its Selective Recognition of Imazethapyr in Soil Samples

[4] doi.org/10.1039/d0ay02116d—Highly selective enrichment and direct determination of imazethapyr residues from milk using magnetic solid-phase extraction based on restricted-access molecularly imprinted polymers

[5] doi.org/10.1002/jssc.201400960—Molecularly imprinted polymer as a novel solid-phase microextraction coating for the selective enrichment of trace imidazolinones in rice, peanut, and soil

Table **S3**: Influence of assay conditions to imazamox recovery rate (%) at IPs modified microplate surface

	Recovery rate, %	
	BSA based IPs	GOx based IPs
IPs dilution (pH 8, blocking buffer 0.5 % wt. BSA)		
10	31±7	14±8
20	64±6	38±4
40	90±3	70±5
80	96±12	86±10
120	93±14	92±15
IPs immobilization pH (IPs 40, blocking buffer 0.5 % wt. BSA)*		
3	71±4	nd
7.4	82±2	nd
9	89±1	nd
Blocking buffer containing BSA, % wt (IPs 40, pH 9) **		
0	99±15	nd
0.1	91±2	nd
0.5	87±3	nd
1	80±2	nd

nd—Negative result, *—

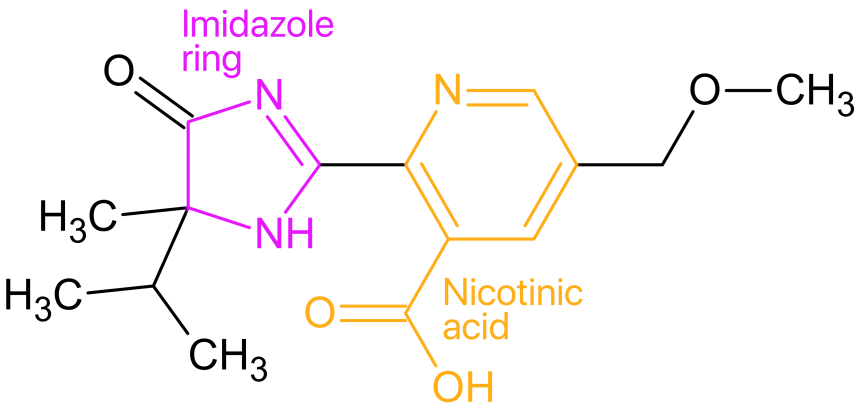

Figure **S1**: Imazamox structure

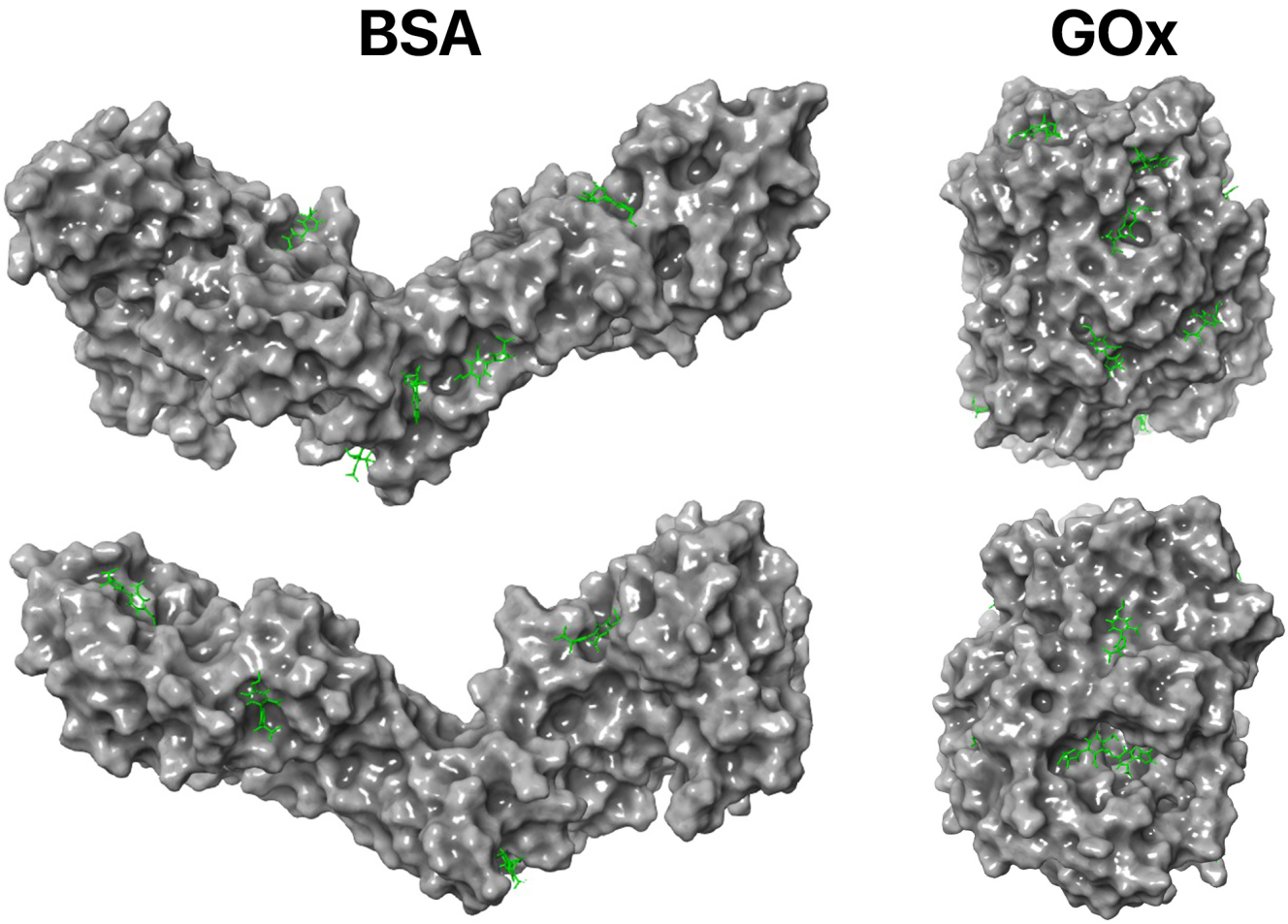

Figure **S2**: Complex protein–imazamox at pH 3.0